Unit-4 Graph Theory :-

A graph is a pair (V, E) where V is a non-empty set and E is a set of unordered pairs of elements taken from the set V.

Here the elements of V are called vertices (points or nodes) and the elements of E are called undirected edges or just edges. The set V is called the vertex set and the set E is called the edge set.

∴ A graph $G(V, E)$ is also denoted as $G = (V, E)$ or $G = G(V, E)$ or just G.

Types of graphs :-

null graph :- A graph which does not contain edges are called null graph or trivial graph.

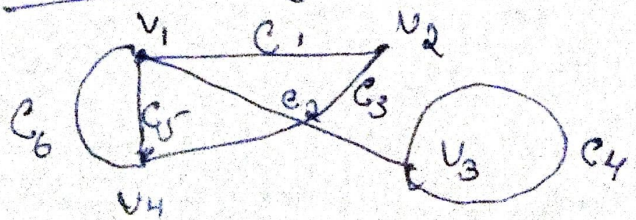
↓ A null graph with only one vertex.

Ex : • A

• B • C

A graph with finite number of vertices and edges are called finite graph. Otherwise it is called infinite graph.

Consider the graph :-

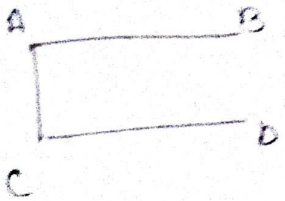


An edge which starts and terminates at the same vertex is called loop. (e_4). If two or more edges have the same vertices are called Parallel edges. Edges e_5 and e_6 have the same end vertices

Here there are Parallel edges

If in a graph there are two or more edges with the same end vertices, the edges are called multiple edges

Simple graph: A graph which contains neither loops or multiple edges is called simple graph.



Multigraph:- A graph which contains multiple edges but no loops are called multigraph.

A graph which contains does not contain loops are called loop-free graph.

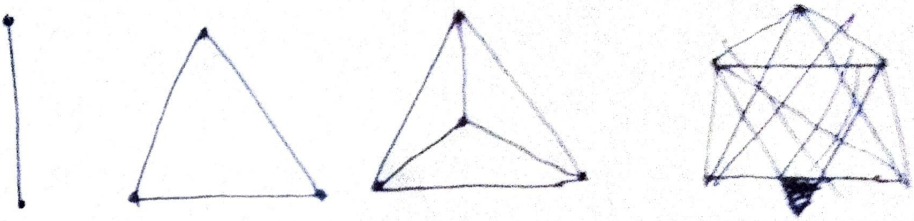
General graph:- A graph which contains loops or multiple edges are called General graph. (or both).

Incidence:- When a vertex v of a graph G is an end vertex of an edge e of the graph G, then that edge e is said to be incident to the vertex v.

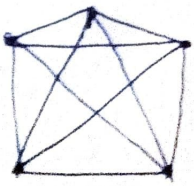
Two non-Parallel edges are said to be adjacent edges if they incident on a common vertex.

* Complete graphs with two, three, four and five vertices are shown below.

* Complete graph with five vertices (K_5). This graph is called the Kuratowski's first graph.



K_2 K_3 K_4

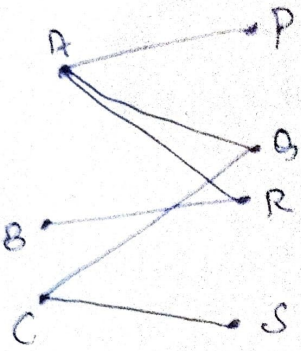


K_5

Bipartite graph:-

Suppose a simple graph G is such that its vertex set V is the union of two of its mutually disjoint nonempty subsets V_1 and V_2 which are such that each edge in G joins a vertex in V_1 and a vertex in V_2. Then G is called a bipartite graph.

If E is the edge set of this graph, the graph is denoted $G = (V_1, V_2; E)$ or $G = G(V_1, V_2)$. The sets V_1 and V_2 are called bipartites (or Partitions) of the vertex set V



Consider Vertex set $V = \{A, B, C, P, Q, R, S\}$

and edge set is $E = \{AP, AQ, AR, BR, CR, CS\}$

The set V is the union of two its subsets $V_1 = \{A, B, C\}$ and $V_2 = \{P, Q, R, S\}$ which are such that (i) V_1 and V_2 are disjoint.

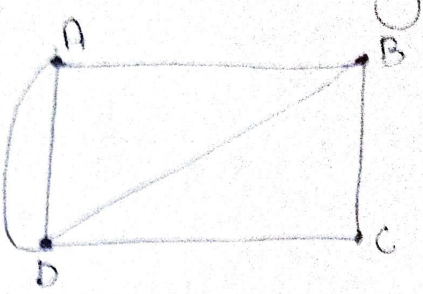
ii) Every edge in G_1 joins a vertex in V_1 and a vertex in V_2.

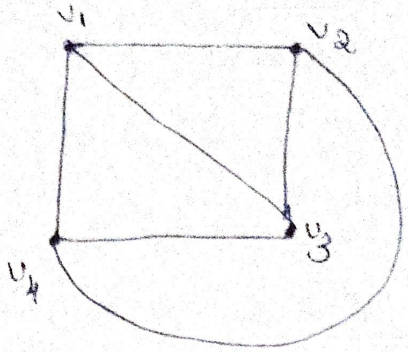
iii) G_1 contains no edge that joins two vertices both of which are in V_1 or V_2. This graph is bipartite graph with $V_1 = \{A, B, C\}$ and $V_2 = \{P, Q, R, S\}$ as bipartites.

Complete Bipartite graph

A bipartite graph $G_1 = (V_1, V_2; E)$ is called a Complete bipartite graph if there is an edge between every vertex in V_1 and every vertex in V_2.

(1) Which of the following is a complete graph?





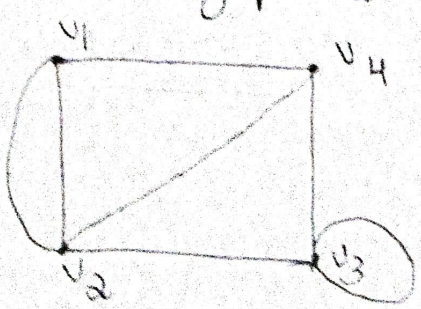
Soln:- The first graph is not complete. It is not simple and there is no edge b/w A and C.

The second graph is complete. It is a simple graph and there is an edge between every pair of vertices.

Vertex degree and Handshaking Property.

Let $G = (V, E)$ be a graph and v be a vertex of G. Then, the number of edges of G that are incident on v with the loops counted twice is called the degree of the vertex v and is denoted by $\deg(v)$ or $d(v)$.

The degrees of the vertices of a graph arranged in non-decreasing order is called the degree sequence of the graph. Also, the minimum of the degrees of vertices of a graph is called the degree of the graph.



$\deg(v_1) = 3$
$\deg(v_2) = 4$
$\deg(v_3) = 4$
$\deg(v_4) = 3$

Isolated Vertex

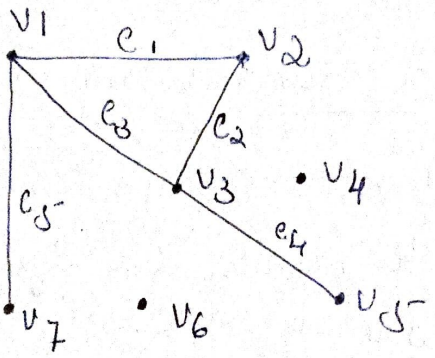
A vertex in a graph which is not an end vertex of any edge of the graph is called an isolated vertex.

A vertex is an isolated vertex if and only if its degree is zero.

Pendant Vertex:- A vertex of degree 1.

An edge incident on a Pendant vertex is called a Pendant edge.

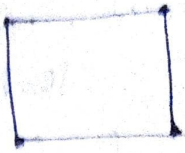
Ex:-



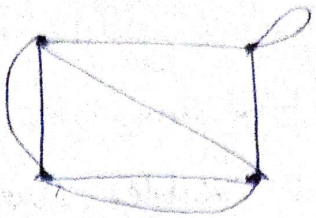
Here the vertex v_4, v_6 are isolated vertices, v_5 and v_7 are Pendant vertices. e_4 and e_5 are Pendant edges.

Regular graph:- A graph in which all the vertices are of the same degree k is called a regular graph of degree k, or a k-regular graph.

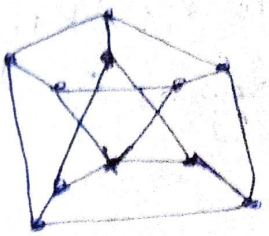
In Particular, a 3-regular graph is called a Cubic graph.



2 regular graph

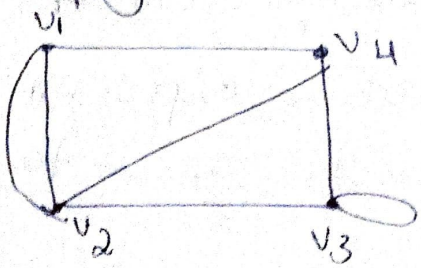


4-regular graph.



* A graph which contains 10 vertices and 15 edges is called the Petersen graph.

Handshaking Property:



$d(v_1) = 3$, $d(v_2) = 4$ $d(v_3) = 4$. $d(v_4) = 3$

Also, the graph has 7 edges

$$\deg(v_1) + \deg(v_2) + \deg(v_3) + \deg(v_4) = 14 = 2 \times 7$$

Property:- The sum of the degrees of all the vertices in a graph is an even number, and this number is equal to twice the number of edges in the graph.

For graph $G = (V, E)$

$$\sum_{v \in V} \deg(v) = 2|E|$$

Theorem: In every graph, the number of vertices of odd degrees is even.

Proof: Consider a graph with n vertices. Suppose k of these vertices are of odd degree so that the remaining $n-k$ vertices are of even degree.

Denote the vertices with odd degree by $v_1, v_2, v_3 \ldots v_k$ and the vertices with even degree by $v_{k+1}, v_{k+2}, \ldots v_n$. Then the sum of the degrees of the vertices is

$$\sum_{i=1}^{n} \deg(v_i) = \sum_{i=1}^{k} \deg(v_i) + \sum_{i=k+1}^{n} \deg(v_i)$$

From the handshaking Property, sum on the left hand side of the above expression is Equal to twice the number of edges in the graph.

$\therefore$ This sum is even.

The second sum is the right hand side is the sum of the degrees of vertices with even degrees

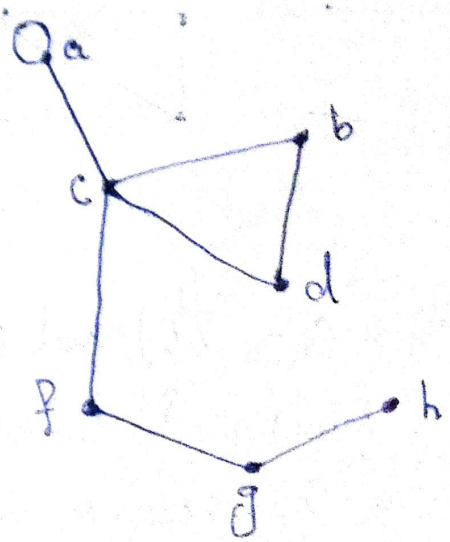
$\therefore$ This sum is also even.

$$\therefore \deg(v_1) + \deg(v_2) + \ldots + \deg(v_k) = \text{even.} \quad \rightarrow (2)$$

$\therefore$ each of $\deg(v_1), \deg(v_2) \ldots \deg(v_k)$ is odd.

$\therefore$ The number of terms in the left hand side of (2) must be even; that is k is even.

① For the graph shown indicate the degree of each vertex and verify the handshaking Property.



Soln: degrees of its vertices are

$\deg(a) = 3$, $\deg(b) = 2$ $\deg(c) = 4$

$\deg(d) = 2$ $\deg(e) = 0$ $\deg(f) = 2$

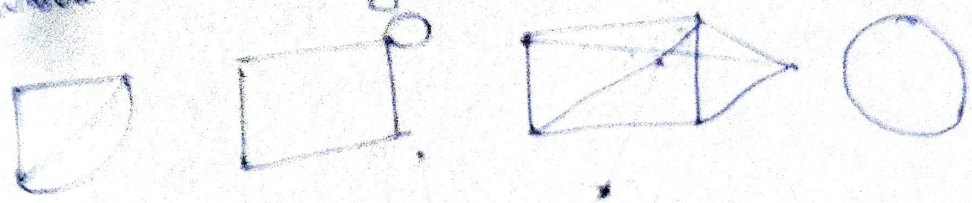
$\deg(g) = 2$ $\deg(h) = 1$

Here e is an isolated vertex and h is a pendant vertex.

∴ The sum of the degrees of vertices is Equal to 16, and it has 8 edges

The sum of the degrees of vertex is equal to twice the number of edges by

Which of the following are complete graph.



Isomorphism

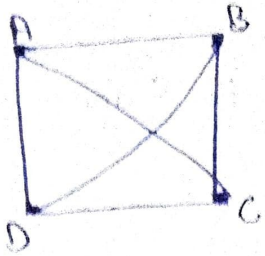
Consider two graphs $G = (V, E)$ and $G' = (V', E')$.
Suppose there exists a function $f: V \to V'$ such that
i) f is a one-to-one correspondence and.
ii) For all vertices A, B of G $\{A, B\}$ is an edge of G if and only if $\{f(A), f(B)\}$ is an edge of G'.
Then f is called an isomorphism between G and G', and we say that G and G' are isomorphic graphs.

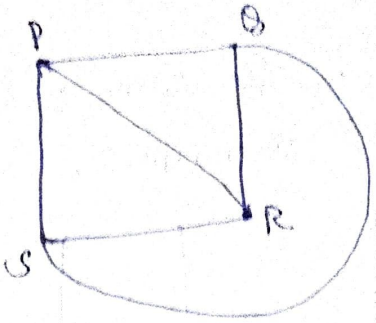
Or

Two graphs G and G' are said to be isomorphic if there is a one-to-one correspondence between their vertices and blw their edges such that the adjacency of vertices is Preserved.

* When G and G' are isomorphic, we write $G \cong G'$

(1)





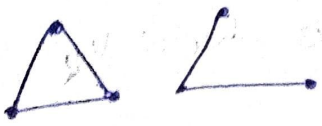
Soln:- Consider the following one-to-one correspondence b/w the vertices of these two graphs.

$A \leftrightarrow P$, $B \leftrightarrow Q$, $C \leftrightarrow R$, $D \leftrightarrow S$

Under this correspondence, the edges in the two graph correspond with each other.

$\{A,B\} \leftrightarrow \{P,Q\}$, $\{A,C\} \leftrightarrow \{P,R\}$,
$\{A,D\} \leftrightarrow \{P,S\}$, $\{B,C\} \leftrightarrow \{Q,R\}$
$\{B,D\} \leftrightarrow \{Q,S\}$. $\{C,D\} \leftrightarrow \{R,S\}$

(2)



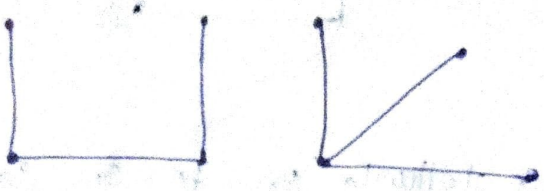
NOTE:- if two graphs are isomorphic then they must have

1. The same number of vertices.
2. The same number of edges.
3. An equal number of vertices with a given degree.

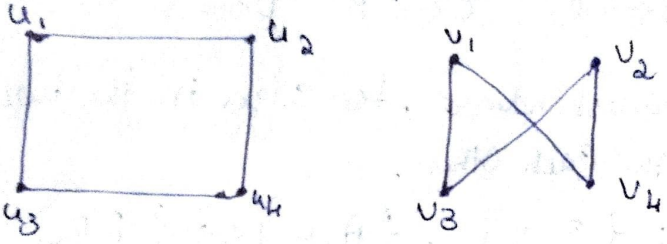
These conditions are necessary but not sufficient.

Two graphs of the same order and the same size need not be isomorphic.

Ex:-



(1) Prove that the two graphs shown below are isomorphic.



Soln:- Now we observe that both graphs have four vertices and four edges.

One-to-one correspondence b/w the vertices of the graph.

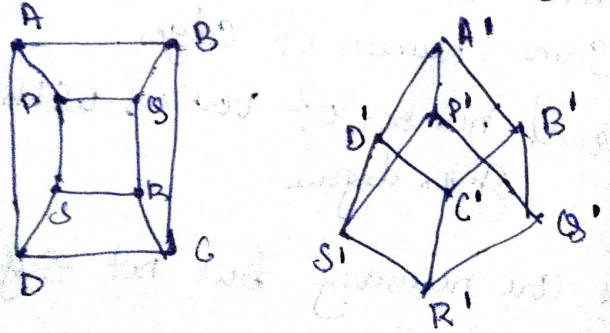
$u_1 \leftrightarrow v_1$, $u_2 \leftrightarrow v_4$, $u_3 \leftrightarrow v_3$, $u_4 \leftrightarrow v_2$

Correspondence b/w the edges.

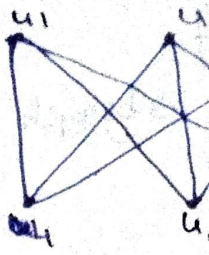
$\{u_1, u_2\} \leftrightarrow \{v_1, v_4\}$ $\{u_1, u_3\} \leftrightarrow \{v_1, v_3\}$

$\{u_3, u_4\} \leftrightarrow \{v_4, v_2\}$ $\{u_3, u_4\} \leftrightarrow \{v_3, v_2\}$

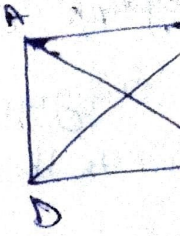
(2)



(3) Show that the



(4) Show that t



Subgraphs:-

Given two

is a subgraph

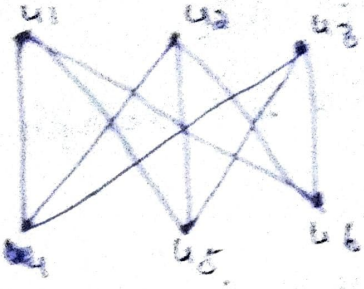
(1) All the

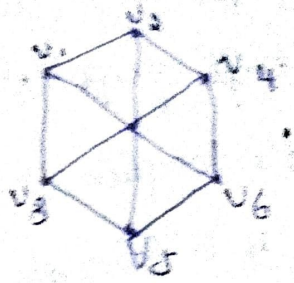
(2) Each

Gr as in G1

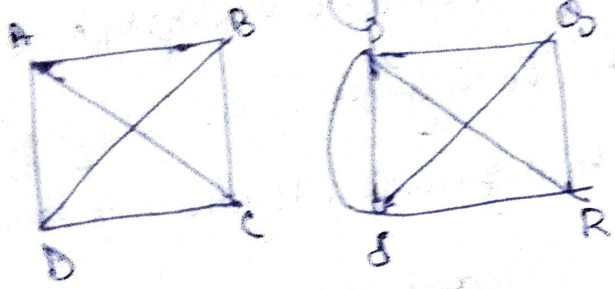
G1

(3) Show that the following two graphs are isomorphic





(4) Show that the following graphs are not isomorphic

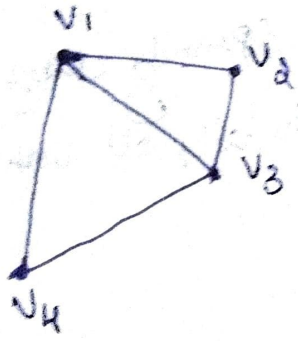


Subgraphs:-

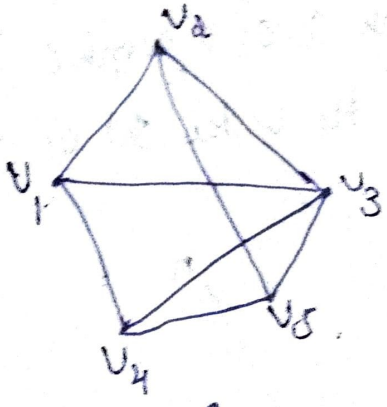
Given two ~~sub~~ graphs G and G_1, we say that G_1 is a subgraph of G if the following conditions hold.

(1) All the vertices and all the edges of G_1 are in G.

(2) Each edge of G_1 has the same end vertices in G as in G_1.



G_1



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Consequences of the definition of a Subgraph

1) Every graph is a subgraph of itself.
2) Every simple graph of n vertices is a subgraph of the complete graph K_n.
3) If G_1 is a subgraph of a graph G_2 and G_2 is a subgraph of a graph G, then G_1 is a subgraph of G.
4) A single vertex in a graph G is a subgraph of G.
5) A single edge in a graph G, together with its end vertices is a subgraph of G.

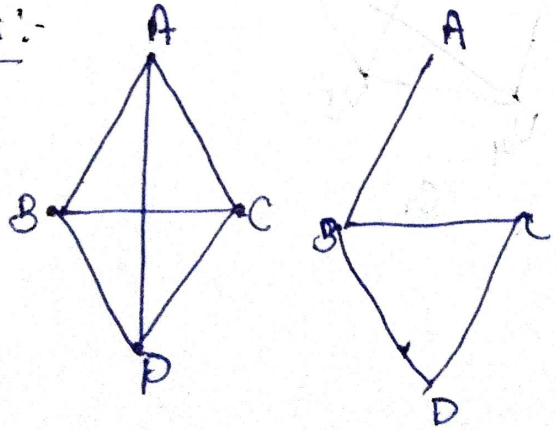
Spanning Subgraph:

Given a graph $G = (V, E)$ if there is a subgraph $G_1 = (V_1, E_1)$ of G such that $V_1 = V$, then G_1 is called a spanning subgraph of G.

or.

A subgraph G_1 of a graph G is a spanning subgraph of G whenever the vertex set of G_1 contains all vertices of G.

Ex:-



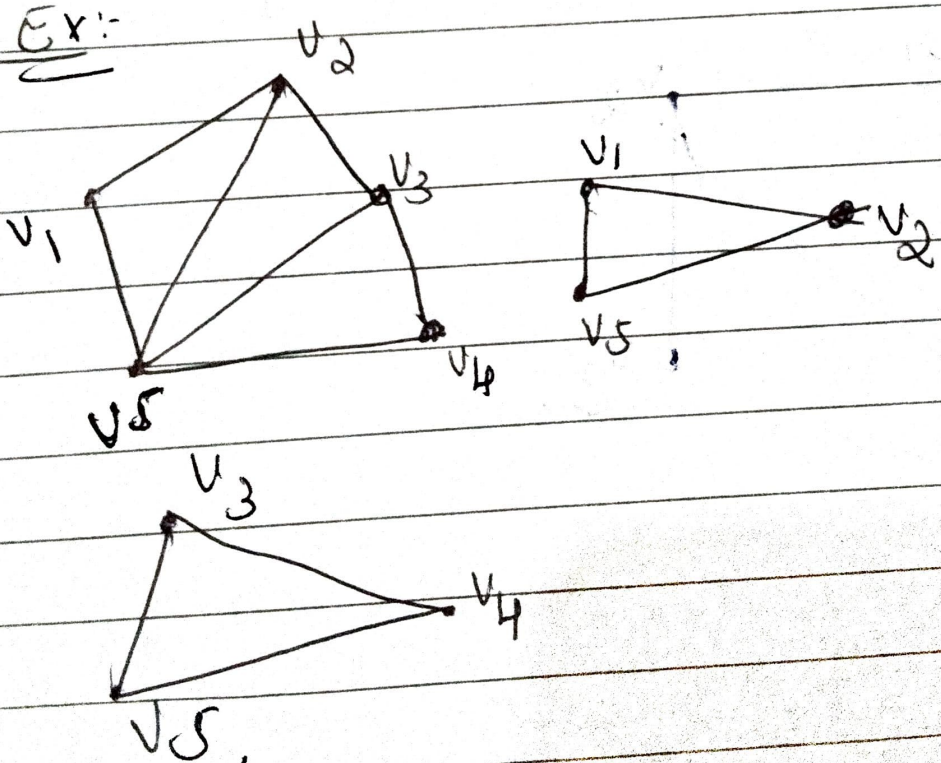
Edge - disjoint and Vertex disjoint Subgraphs

Let G be a graph and G_1 and G_2 be two subgraphs of G. Then:

(1) G_1 and G_2 are said to be edge - disjoint if they do not have any common edge.

(2) G_1 and G_2 are said to be vertex disjoint if they do not have any common edge and any common vertex.

Ex:-



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Walks and their Classification

Here we Consider five important subgraphs of a graph, called a walk, a trail, a circuit, a Path and a cycle.

Walk:-

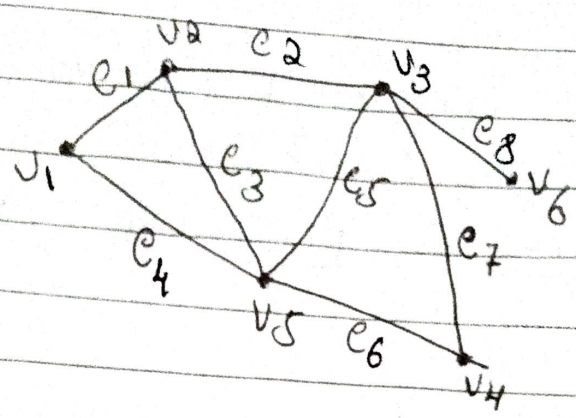
Let G be a graph having at least one edge. In G Consider a finite, alternating sequence of vertices and edges of the form.

$$v_i e_j v_{i+1} e_{j+1} v_{i+2} \ldots, e_k v_m.$$

which begins and ends with vertices and which is such that each edge in the sequence is incident on the vertices Preceding. Such sequence is called a walk in G. In a walk, a vertex or an edge (or both) Can appear more than once.

* The number of edges present in a walk is called its length.

Ex:-



(i) The sequence $v_1 e_1 v_2 e_2 v_3 e_8 v_6$ is a walk of length 3. And no vertex & no edge is repeated.

(ii) The sequence $v_1 e_4 v_5 e_3 v_2 e_2 v_3 e_5 v_5 e_6 v_4$ is a walk of length 5.
Vertex v_5 is repeated but no edge is repeated

(iii) The sequence $v_1 e_1 v_2 e_3 v_5 e_3 v_2 e_2$ is a walk of length 4.
The edge e_3 is repeated and vertex v_2 is repeated.

The vertex with which a walk begins is called the initial vertex (or the origin) of the walk and the vertex with which a walk ends is called the final vertex (or the terminal)

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* The initial vertex and the final vertex of a walk are together called its (terminal vertices)
* The terminal vertices need not be distinct.
* non terminal vertices of a walk are called its internal vertices.

→ A walk having u as the initial vertex and v as the final vertex is called a walk from u to v or u-v walk.

→ A walk that begins and ends at the same vertex is called a closed walk.
→ A closed walk is a walk in which the terminal vertices are coincident.
→ A walk which is not closed is called an open walk.
→ an open walk is a walk that begins and ends at two different vertices.

Ex:- $v_1 e_1 v_2 e_3 v_5 e_4 v_1$ is a closed walk and $v_1 e_1 v_2 e_2 v_3 e_5 v_5$ is an open walk.

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Trail and Circuit :-

→ If in an open walk no edge appears more than once, then the walk is called a trail.

→ A closed walk in which no edge appears more than once is called a circuit

Ex :- The open walk $v_1 e_1 v_2 e_3 v_5 e_3 v_2 e_2 v_3$ is not a trail (because edge e_3 is repeated).

Where as $v_1 e_4 v_5 e_3 v_2 e_2 v_3 e_5 v_5 e_6 v_4$ is a trail.

Closed walk $v_1 e_1 v_2 e_3 v_5 e_3 v_3 e_2 v_2 e_5 v_5 e_4 v_1$.
is not a circuit, (because e_3 is repeated)

Where as the closed walk $v_1 e_1 v_2 e_3 v_5 e_5 v_3 e_7 v_4 e_6 v_5 e_4 v_1$ is a circuit

Path and Cycle

→ A trail in which no vertex appears more than once is called a Path.

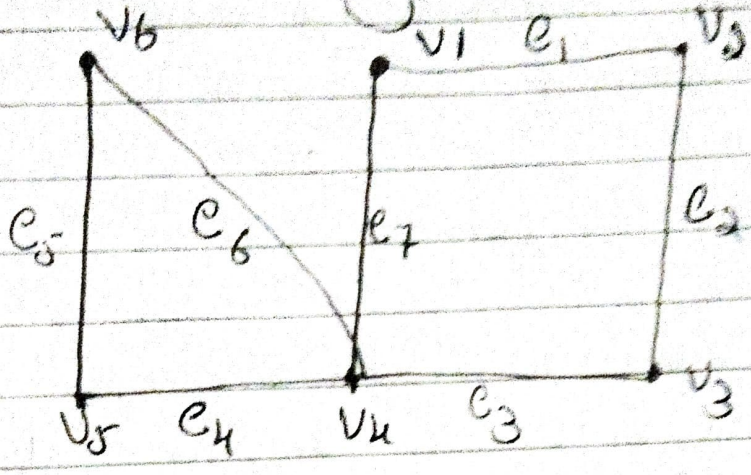
→ A Circuit in which the terminal vertex does not appear as an internal vertex (also) and no internal vertex is repeated is called a Cycle.

Ex:-

The trail $v_1 e_1 v_2 e_3 v_5 e_5 v_3 e_7 v_4$ is a Path whereas the trail $v_1 e_4 v_5 e_3 v_2 e_2 v_3 e_5 v_5 e_6 v_4$ is not a Path (because v_5 appear twice).

The Circuit $v_2 e_2 v_3 e_5 v_5 e_3 v_2$ is a cycle whereas the Circuit $v_2 e_1 v_1 e_4 v_5 e_5 v_3 e_7 v_4 e_6 v_5 e_3 v_2$ is not a cycle. (v_5 appear twice)

① For the graph shown, indicate the nature of the following walks.



(i) $v_1 e_1 v_2 e_2 v_3 e_2 v_2$

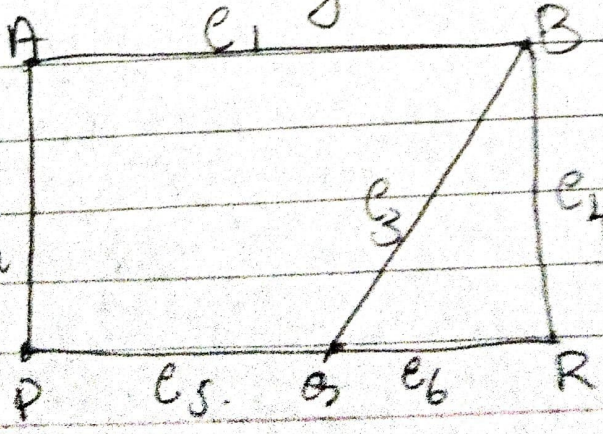
(ii) $v_4 e_7 v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5$

(iii) $v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5$

(iv) $v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_7 v_1$

(v) $v_6 e_5 v_5 e_4 v_4 e_3 v_3 e_2 v_2 e_1 v_1 e_7 v_4 e_6 v_6$.

② Consider the graph shown below. Find all Paths from vertex A to vertex R and indicate their lengths.



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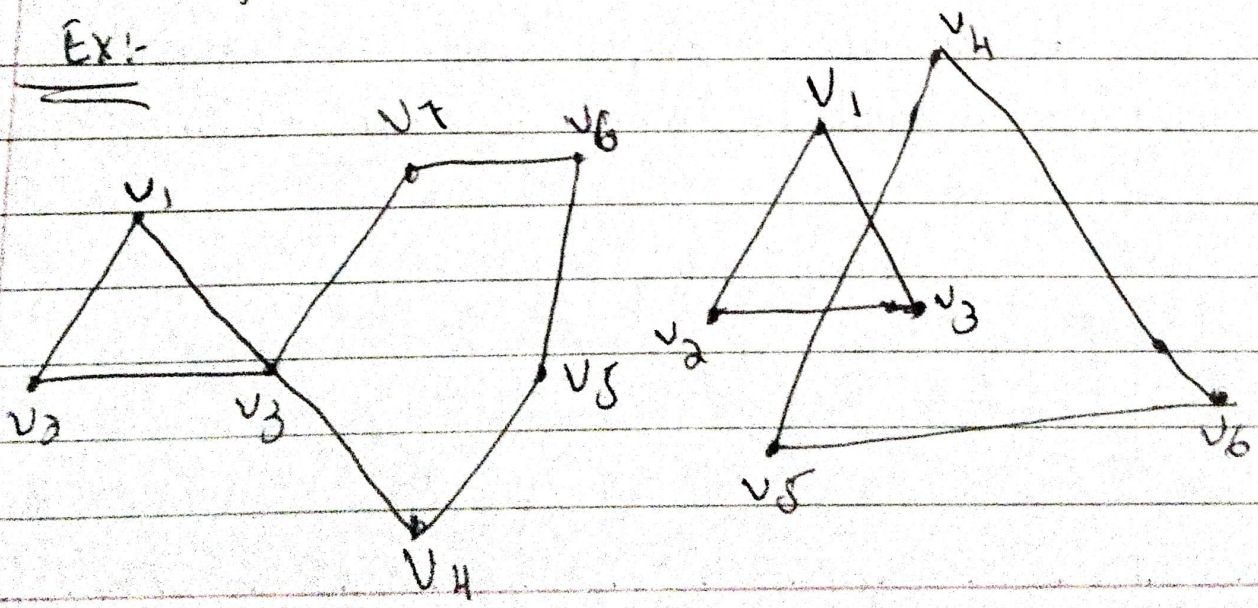
Connected and Disconnected graphs

Consider a graph G of order greater than or equal to two. Two vertices in G are said to be connected if there is at least one path from one vertex to the other.

→ G is connected if every pair of distinct vertices in G are connected. Otherwise G is called a disconnected graph.

A graph G is said to be (i) connected if there is at least one path between every two distinct vertices in G, and.
(ii) disconnected if G has at least one pair of distinct vertices between which there is no path.

Ex:-



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* All walks and, all trails, all circuits, all Paths and all Cycles in a Graph G are Connected Subgraphs of G.

Euler circuits and Euler trails

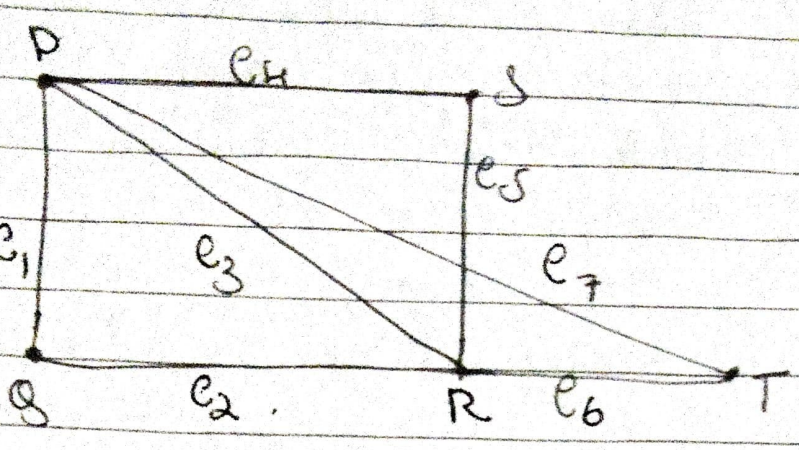
Consider a Connected graph G. If there is a circuit in G that contains all the edges of G, then that circuit is called an Euler circuit in G.

If there is a trail in G that contains all the edges of G, then that trail is called an Euler trail in G.

A Connected graph that contains an Euler circuit is called an Euler graph.

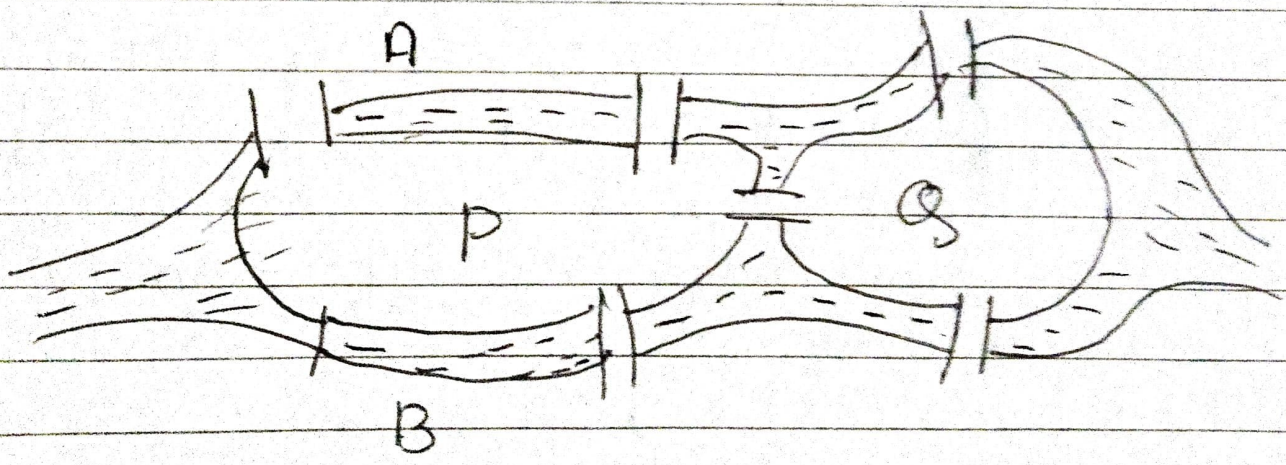
A Connected graph that contains an Euler trail is called a Semi-Euler graph.

Ex:-



$P e_1 Q e_2 R e_3 P e_4 S e_5 R e_6 T e_7 P$
is a closed walk & an Euler Circuit,
∴ It is an Euler graph.

The Konigsberg Bridge Problem :-



In the eighteen Century City named Konigsberg in East Prussia (Europe), there flowed a river named Pregel River which divided the City into four Parts

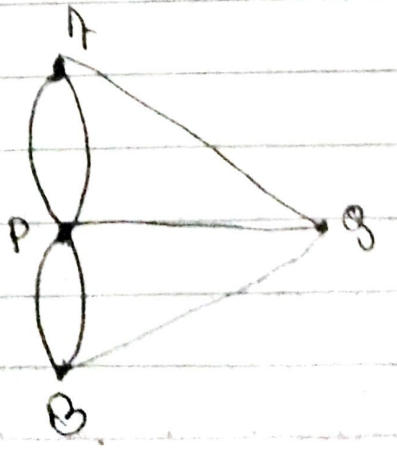
Two of these parts were the banks of the river and two were islands. These parts were connected with each other through seven bridges.

→ By starting at any of the four land areas, can we return to that area after crossing each of the seven bridges exactly once?

This problem is known as "Konigsberg Bridge Problem"

The Land areas of the city by A, B, P, Q, where A, B are the banks of the river and P, Q are the islands.

Construct a graph by treating the four land areas as four vertices and the seven bridges connecting them as seven edges.



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$\deg(A) = \deg(B) = \deg(C) = 3$

$\deg(P) = 5$.

which are not even.

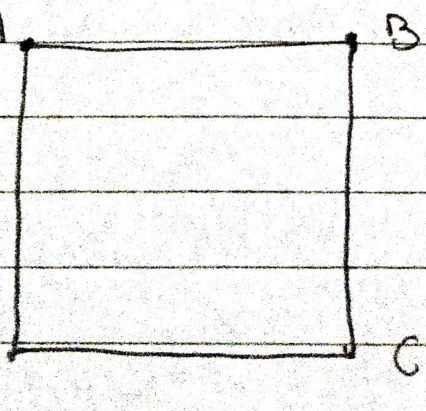
∴ The graph does not have an Euler circuit. This means that there does not exist a closed walk that contains all the edges exactly once.

Hamilton Cycles and Hamilton paths

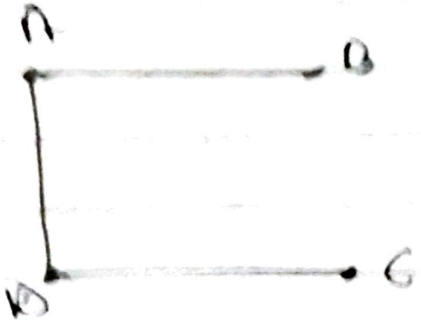
Let G be a connected graph. If there is a cycle in G that contains all the vertices of G, then that cycle is called a Hamilton cycle in G.

A graph that contains a Hamilton cycle is called a Hamilton graph.

Ex:



-> A Path in a connected graph which includes every vertex (but not necessarily every edge) of the graph is called a Hamilton / Hamiltonian Path in the graph.

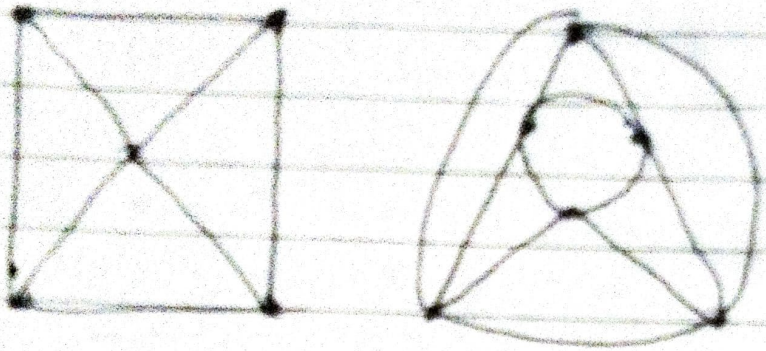


Planar graph.

A graph which can be represented by at least one plane drawing in which the edges meet only at the vertices is called a planar graph.

A graph which cannot be represented by a plane drawing in which the edges meet only at the vertices is called a non-planar graph.

① Show that (i) a graph of order 5 and size 8 and (ii) a graph of order 6 and size 12 are planar graphs.



(a)